

CS 2050 — Notes 09/02

Precedence of quantifiers. $\forall$ and $\exists$ have higher precedence than all logical operators.

Nested quantifiers. A nested quantifier is when one quantifier is within the scope of another.

Example. $\forall x \exists y (x - 2y = 5)$

Everything within the scope of a quantifier can be thought of as a propositional function.

The order of quantifiers is important unless all the quantifiers are the same.

$\exists x \forall y P(x, y)$ is not logically equivalent to $\forall y \exists x P(x, y)$.

Definition. $P \equiv Q$ means $P \leftrightarrow Q$ is a tautology.

Example. $\forall x (W(x) \rightarrow F(x))$

For all people x , if x is a wizard, then x has a funny hat. All wizards have a hat.

Example. $\forall x (W(x) \wedge F(x))$

For all people x , x is a wizard and has a funny hat. Everyone is a wizard and has a funny hat.

Example. $\exists x (W(x) \rightarrow F(x))$

For some person x , if x is a wizard, then x has a funny hat.

Example. There is an adventurer in Faerun who could use a bow and a sword.

$$\exists x (P(x) \wedge Q(x))$$

Example. There is an adventurer in Faerun who could use a bow but could not use a sword.

$$\exists x (P(x) \wedge \neg Q(x))$$

Example. Every adventurer in Faerun could use a bow or use a sword.

$$\forall x (P(x) \vee Q(x))$$

Example. Some student in this class has played Dungeons and Dragons.

$P(x) = x$ has played D&D. Domain = students in the class.

$$\exists x P(x)$$

Example. $\forall x \exists y (x < y)$

For all x , there exists a real number y such that $x < y$.

Example. $\exists x \forall y (x < y)$

There exists x such that for all y , $x < y$.

Example. $\forall x \forall y ((x < 0) \wedge (y < 0)) \rightarrow (xy > 0)$

For every x and every y , if $x < 0$ and $y < 0$, then $xy > 0$.